Sardar Abdul Moiz Khan

Software Engineer

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Experience

- **KNYSYS** *Dec 2025 – Present*
  Jr. Software Developer Karachi, Pakistan
  - Led the architecture and development of the UN Satellite Operator Management System on Next.js, accelerating deployment cycles through GitHub Actions CI/CD while ensuring secure machine-to-machine communication through robust X-API key management.
  - Worked on Multi-tenant SAAS telehealth system, handling maintenance and integrations with Intercom, Junction Labs, and Stripe .
  - Designed a configurable Risk Analysis and Audit Trail System covering **90%** of declaration types, with automated alerts, end-to-end traceability, audit logging, and a feedback loop to continuously improve accuracy.
  - Developed a customizable audit workflow using Prefect, with configurable settings, dynamic task assignments, and escalation mechanisms. Integrated Textract, VoyageAI, Anthropic, LangGraph, LangSmith, and Sentry to automate business processes, verify declarations, and improve error handling.

- **InfraNext SMC-Pvt. Ltd.** *August 2025 – Oct 2025*
  Full Stack Developer Karachi, Pakistan
  - Improved database query performance by restructuring SQL joins and relationships with caching, cutting average API response times from **450ms** to under **200ms**.
  - Engineered an audit logging framework, and integrated Stripe and Shopify via REST/GraphQL pipelines to deliver multi-tenant onboarding, segregated billing workflows, and end-to-end order synchronization on Laravel Backend.

- **NCAI Smart City Lab, NEDUET** *Jun 2024 – Nov 2024*
  AI Intern Karachi, Pakistan
  - Trained computer vision models for driver drowsiness, focus detection, and road segmentation, while scraping and collecting over **10,000+ road sign images** for sign detection. Also, integrated YOLO-based speed and distance estimation for an autonomous vehicle's autopilot and automatic-braking systems.
  - Implemented an automated pipeline to transform NoSQL soil data into a structured SQL format, integrating distributed FPGA devices to provide real-time soil insights. Developed a dashboard supporting 12-month historical analysis with **sub-3-second** query performance using Folium and FastAPI.

Education

- **NED University of Engineering and Technology** *Aug 2022 – June 2026*
  BE in Computer Systems Engineering Karachi, Pakistan
  - CGPA: **3.835/4.00**

Skills

Soft Skills: Communication, Leadership, Agile Development, Problem Solving, Adaptability.

Technical Skills: Python, React, Next.js, TypeScript, FastAPI, LLM Integration, RAG, LangGraph, LangSmith, Observability, Prompt Engineering, Vector Databases, Supabase, PyTorch, Deep Learning, Docker, Redis, Kafka, Database Design, Sentry.

Projects

- **Reiseagent: AI Travel Assistant Platform**
  Tools: .NET 9, C#, PostgreSQL (pgvector), Azure AI Search, OpenAI, Next.js, SignalR, Docker [ ]
  - Built a production RAG platform integrating OpenAI-based generation, real-time streaming, and dialogue orchestration for personalized, context-aware travel recommendations.
  - Engineered a hybrid retrieval pipeline (pgvector semantic embeddings + full-text search + Azure AI Search fallback), with a reliability "Quality Gate" layer that caught and prevented zero-result and low-confidence failures before they reached users.

- Designed and ran a 29+ scenario evaluation framework covering multilingual queries, typo handling, ambiguous intent, and seasonal constraints; containerized the full platform with Docker, automated migrations, and seeded test datasets for repeatable evaluation runs.

- **WebMAve: Real-Time Video Calling Platform**
  Tools: Next.js, TypeScript, Stream Video SDK, Clerk, Supabase, Tailwind CSS, Vercel [|]
  - Built a complete video meeting workflow with instant calls, scheduled meetings, invite-link joining, and dedicated Upcoming, Previous, and Recordings views for better call lifecycle management.
  - Added Audience Mode in meeting rooms, where attendees can focus on the host stream while preserving a separate host view and controls, improving large-session usability.
  - Developed real-time in-call session messaging with Supabase Realtime, including live updates, message history loading, auto-scroll behavior, and new-message indicators in the meeting UI.

- **Argus PM: Engineering Intelligence & Project Management Platform**
  Tools: Python, ReactJS, Supabase, FastAPI, PydanticAI, DeepSeek, LangGraph, LangSmith
  - Architected a multi-agent organizational assistant using LangGraph, PydanticAI and FastAPI, coordinating tool use, memory, and autonomous retrieval to deliver evidence-based answers through email.
  - Built API integrations for GitHub, Jira and Slack to centralize engineering activity and team data, powering hybrid search and LLM-driven insights with **DeepSeek**. Used LangSmith for LLM tracing and monitoring across agent workflows.
  - Implemented secure multi-tenant architecture (JWT authentication, RBAC, team-level data isolation, encrypted credentials) with a React dashboard tracking PRs, issues, metrics, and alerts.

- **DentAI: Dental X-Ray Segmentation Web-App**
  Tools: YOLOv5, FastAPI, React, Python, Custom Dataset [ ]
  - Analyzed and cleared the Dental X-Ray Dataset, then trained a custom YOLOv5 Segmentation Model achieving an **F1 score of 70%** for Dental Anomaly Detection.
  - Built a high-throughput and responsive backend using FastAPI with concurrent threading and buffering, integrated with a React frontend for real-time diagnosis.

- **Armour Security: MDM Platform**
  Tools: Python, Flutter, Kotlin, Firebase, Docker, VPS [ ]
  - Developed a full-scale B2B Mobile Device Management (MDM) platform for centralized enterprise device monitoring and management, with advanced security features including remote lock/unlock, live location tracking, SIM change detection, unauthorized reset protection, and USB reset prevention to enhance organizational device security and compliance.
  - Built and deployed scalable backend infrastructure using Docker, and VPS hosting, enabling secure real-time synchronization, efficient device communication, and reliable large-scale operations.

Certifications

- **HEC Generative AI Program**
  Higher Education Commission (HEC) Pakistan & Pak Angels
- **Machine Learning Specialization**
  Stanford University & DeepLearning.AI — *Coursera*
- **Convolutional Neural Networks**
  Stanford University & DeepLearning.AI — *Coursera*
- **Applied Data Science Lab**
  WorldQuant University — *Credly*

Additional Information

Achievement: Secured **99th Percentile** in the National Skill Competency Test.
Languages: Urdu (Native), English (Proficient)
Interests: Badminton (Occasionally), Tinkering with Electronics, Video Games, Puzzle Solving.